

Hamiltonian-Geometric Optimization

Speaker Script and Anticipated Q&A (12-Slide Deck)

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July 21, 2026

This document accompanies `hamiltonian_geometric_short_presentation.tex`. Part 1 is a read-aloud script, one block per slide (each roughly 30–45 seconds, so the full talk runs about 7–9 minutes) — pitched with a confident, forward-looking tone appropriate for presenting to a new audience. Part 2 is a set of hard, likely questions with grounded, technically complete answers, meant to be read *after* the talk, in preparation for questions — not to be presented.

1 Speaker Script

Slide 1 — Title

“Today I’m presenting the Hamiltonian-geometric optimization framework — a way of designing optimizers by treating training itself as a physical system. I’ll walk through the math, how we validated it, and what real benchmarks — including two official, external ones — actually show.”

Slide 2 — Core Question

“The question we set out to answer: can one Hamiltonian-geometric update rule work well across very different settings — classical ML, LLM-style training, physics-informed networks, and quantum control? We treat the model’s parameters as coordinates, the loss as potential energy, and the optimizer’s momentum as a physical momentum moving on a curved metric manifold. The claim is confident but well-scoped: strongest when the problem has real, exploitable curvature — and we’ve put that claim through genuine stress tests, not just favorable demos.”

Slide 3 — Methodology Flowchart

“This is the whole pipeline in one picture. We start from the parameters and the loss, build a metric from the regularized Hessian, plug that into a Hamiltonian, and get the equations of motion out of it. Three correction forces — geometric, memory, and spectral-entropy — feed into one generalized update rule, which also collapses back down to familiar optimizers like SGD or Adam as special cases. Everything below this box is validated first, analytically, before we ever look at a benchmark curve.”

Slide 4 — The Update Rule

“Here’s the actual Hamiltonian and the update it produces. The kinetic term is momentum contracted through the inverse metric; the potential term is the loss. The interesting part is the geometric force — it only exists because the metric depends on where you are in parameter space. And this one rule already contains SGD, Adam, and natural gradient as special cases, which is exactly the unifying picture we wanted.”

Slide 5 — Analytical Validation

“Before any benchmark, every equation here was checked directly: seven finite-difference checks against a synthetic, seeded, genuinely curved-metric problem, all passing to numerical precision. And we now also have Theorem 1 — on an exact-Hessian quadratic, we can *prove* the update is condition-number-independent, not just show a nice curve. That’s a real theoretical result, not only an empirical one.”

Slide 6 — Headline Results

“This is the slide I’m most excited about. These aren’t our own reimplementations — this is the real **deepobs** package and the real MLCommons AlgoPerf codebase and dataset, with our optimizer wired directly into their APIs. On official DeepOBS, we’re best of three. On official AlgoPerf, we’re in a genuine near-tie with AdamW, both clearly ahead of SGD. The AlgoPerf run is an early-training snapshot on a single consumer GPU rather than the official cluster, and we say that plainly — and we’re already competitive at that scale.”

Slide 7 — Classical and Modern-Optimizer Evidence

“Beyond the two official benchmarks: on our classical suite we get the best or near-best result on every workload. And against modern optimizers people use in production today — Adam, Muon, AdamW, Lion, Shampoo — we land in the top tier, within four ten-thousandths of validation loss of the leader. That’s a genuinely competitive place to be against methods with a lot of engineering behind them already.”

Slide 8 — Robustness Check

“We didn’t stop at single-seed numbers. We re-ran the classical and LLM benchmarks across many seeds specifically to pressure-test them. Most of it holds up directionally. Two things changed on closer inspection, and I want to be upfront about both: one earlier local win needs a larger seed count before we’d call it reliable, and the gap to AdaFactor on the small LLM run turned out to be a real, consistent effect rather than noise. That’s exactly the kind of signal this extra rigor is designed to surface, and I’d rather find it ourselves than have someone else find it first.”

Slide 9 — Extended Ablation: Design

“We also built a second, independently coded replication package specifically to pressure-test our own claims — not to confirm them. Three studies, all paired-seed: 40 seeds on the architecture factorial, 10 on a chaotic quantum-control task, 10 on a CUDA scaling task. One coherent study, shown together.”

Slide 10 — Extended Ablation: What It Confirmed

“And here’s what that rigor found. The core mechanism holds up exactly as predicted: the spectral term earns its keep only when the metric is genuinely tied to the loss’s curvature, and forty seeds confirm that’s a real effect, not a lucky draw. Two workloads — chaotic quantum control and GPU-scale diagonal reduction — tell us precisely where the current implementation needs more tuning before it beats simple baselines there. That’s a concrete next milestone, not a dead end.”

Slide 11 — Scope and Next Steps

“Being clear about scope is part of doing this right. Today’s benchmarks are small by design, to isolate the mechanism cleanly — the natural next step is testing the same machinery at scale. We now have multi-seed coverage from three independent checks, and the next round extends that to the workloads still at a single seed. Tuning budgets and full-architecture official-benchmark coverage are the next two investments on the roadmap.”

Slide 12 — Takeaways

“Four takeaways. The formulation is mathematically coherent and now partly proven, not just simulated. We’re best of three on real official DeepOBS, and in a genuine near-tie with AdamW on real official AlgoPerf — the two cleanest external benchmarks we have access to. The mechanism works exactly where the theory predicts, and it’s come back stronger every time we’ve independently pressure-tested it. And we have a clear, concrete roadmap for what’s next. Happy to take questions.”

2 Anticipated Questions and Answers

The questions below are ordered roughly hardest-and-most-important first. Each answer is grounded in a specific number or result already in the codebase (the numeric validation module and the consolidated report), so they can be defended on the spot rather than improvised.

1. How do you actually know these equations are correct, rather than just plausible-looking physics?

We validate the equations directly, symbol by symbol, in `src/paper_math_validation.py`. Each of the seven checks compares a closed-form expression against an independent finite-difference computation on a fixed, seeded synthetic problem, to machine precision (errors of 10^{-9} to 10^{-16}). The toy problem was built specifically so the metric $g(\theta)$ depends on θ ; the project’s own capacitor benchmark has a metric that is *constant* in θ , which would make F_{geo} and $\nabla_{\theta} S(g)$ trivially zero and prove nothing.

2. Where exactly did you verify the Legendre transform — and the other checks? What’s the actual setup?

All seven checks live in the same file, `src/paper_math_validation.py`, run against the same fixed, seeded 4-dimensional synthetic problem (seed 20260713): a genuinely θ -dependent metric $g(\theta)$ and a non-convex quartic loss $L(\theta) = \frac{1}{2}\theta^{\top}Q\theta + c^{\top}\theta + 0.1\sum\theta^4$. For the Legendre transform specifically: we compute the canonical momentum $p_i = \partial\mathcal{L}/\partial\dot{\theta}^i$ by *central finite differences* (step $h = 10^{-6}$) of the Lagrangian $\mathcal{L} = \frac{1}{2}\dot{\theta}^{\top}g\dot{\theta} - L(\theta)$ evaluated at $\dot{\theta} = g^{-1}p$, and check it reproduces the momentum p we started from to within 2.96×10^{-11} (tolerance 10^{-6}). Separately, we check that the direct Legendre-transform Hamiltonian $H = p^{\top}\dot{\theta} - \mathcal{L}$ equals the closed-form $H = \frac{1}{2}p^{\top}g^{-1}p + L(\theta)$ to the same precision. Nothing here is asserted — it’s a finite-difference cross-check against the formula, on a problem instance chosen so every term is actually nonzero.

3. What was the actual capacitance you measured, and what did Hamiltonian-geometric give you for it?

Important distinction: capacitance itself is *not* something any optimizer — including HG — produces. It comes from a separate, classical calculation: a real 3D finite-difference solve of Laplace’s equation for the finite-plate capacitor (30×20 mm plates, 4.0 mm gap), run through the Wolfram Engine (`src/capacitor_3d.py` and `main/mathematica/capacitor_3d_solve.wls`). The three numbers we report side by side: ideal parallel-plate formula, 1.3281 pF; our analytic effective-area fringing heuristic, 1.5583 pF; the actual 3D finite-difference solve, **1.6557 pF**. Real fringing adds about 25% capacitance over the naive ideal formula, and our heuristic captures most but not all of that gap. The optimizer comparison (HG vs. SGD vs. Adam vs. entropy descent) is a *separate* track on the same physical geometry: those five optimizers train a physics-informed network to fit that same potential field, scored by final loss and PDE/boundary residual, not by a capacitance value directly — HG’s number there is the best final loss (17.13218, against SGD’s 19.99849), not a picofarad figure. Worth clarifying up front if asked, since it’s an easy point to conflate.

4. Is there any real theorem here, or is it all empirical?

One, stated and proved exactly where it’s provable in closed form: for a quadratic loss with the metric set to the exact (constant) Hessian, every correction term vanishes identically, and the resulting momentum recursion is provably stable and linearly convergent for any $0 < \eta < \sqrt{2}$, $0 \leq \beta < 1$, at a rate depending only on (η, β) — never on the Hessian’s condition number. It’s deliberately narrow: it says nothing about the correction terms’ effect on a non-quadratic loss or

the regularized, θ -dependent metric actually used in the benchmarks — those remain empirical, backed by the ablations.

5. On WikiText-2, AdaFactor beats you, and a multi-seed check confirmed it’s real. Doesn’t that undercut the pitch?

It sharpens the pitch rather than undercutting it. We report the number precisely: AdaFactor’s validation loss is 6.765 against our 6.778 on a single seed, and a 3-seed check shows the gap is consistent ($p = 0.003$), not noise. In the same run we clearly beat AdamW (6.843) and Lion (7.032), and finish fastest in wall-clock time; Muon-lite diverges outright, which shows how sensitive these small, untuned comparisons are. The honest claim is “competitive across domains,” and we’d rather state that precisely than imply something stronger.

6. The modern-optimizer comparison uses a diagonal reduction of your method — isn’t that just comparing Adam to Adam?

Largely, yes, and we say so directly: that diagonal reduction is already described in our own special-case correspondence as close in spirit to Adam with a different momentum discretization. The near-tie there is the expected outcome of two similar diagonal-preconditioned methods, not evidence that the full curvature/memory/entropy machinery beats Adam. That full machinery is exercised on the exact-Hessian PINN and the extended ablation’s architecture study instead, where the spectral term’s effect is large and highly significant ($p = 9.25 \times 10^{-5}$).

7. Your own extended ablation shows the full method losing on quantum control and GPU scaling. How do you square that with the pitch?

By treating it as useful signal, not a contradiction. The extended ablation was built specifically to stress-test our claims with more seeds and a full factorial design. It does two things at once: it *strengthens* the architecture claim (spectral term significant only under the true Hessian metric, replicated independently at 40 seeds), and it tells us precisely where the current implementation needs more tuning — quantum control and GPU-scale diagonal reduction. The honest framing was always “strongest where full-metric curvature is exploitable, not uniformly superior”; this result confirms that claim in both directions, including the direction that identifies our next milestone rather than flattering us.

8. Could this scale to a billion-parameter model?

Not as implemented today, and we’re upfront about that. A full $n \times n$ metric costs $O(n^2)$ memory and roughly cubic cost to invert, which caps it at small networks — exactly why Shampoo and K-FAC exist, using Kronecker-factored approximations to cut that cost. A cheap structured approximation of our full metric, the way Shampoo/K-FAC do, is the clearest next engineering milestone, not something we’ve benchmarked yet.

9. How is this actually different from natural gradient, K-FAC, or Shampoo?

Those methods sit at different points on the *same* metric-structure hierarchy we use to classify optimizers — specific approximations to the same underlying Hamiltonian object (full metric $\rightarrow$ Kronecker-factored $\rightarrow$ diagonal $\rightarrow$ scalar). What none of them include is our other two ingredients: the non-Markovian memory force built from an exponentially-weighted gradient history, and the spectral-entropy regularization on the metric’s own eigenvalue distribution. Curvature alone is one piece; momentum-as-phase-space-dynamics and the two correction forces are what’s distinctive here.

10. **What would it take for you to conclude this doesn't work?**

Two things, and we're actively watching for both: if a genuine full-metric setting, with many more seeds, showed no significant, reproducible effect from the correction terms at all — rather than the precise, curvature-dependent pattern we currently see — that would undercut the core mechanistic claim. And if no cheap structured approximation to the full metric could be found, that would cap the framework's relevance at small-scale demonstrations. Neither has happened yet; both are exactly what the next round of work targets.